DecodeLabs Internships 2026

DEVOPS PROJECT 1

Linux & Command Line Basics

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Submitted To

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Project Details

Project Number: Project 1

Project Title: Linux & Command Line Basics

Technology Used: Linux (WSL Ubuntu), Command Line Interface (CLI)

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Declaration

I hereby declare that this project report titled **"Linux & Command Line Basics"** has been completed by me as part of the DecodeLabs Internships Program. The work presented in this report is based on my own practical implementation and learning experience using Linux commands and command-line operations in a Linux environment

Project 1: Linux & Command Line Basics – Introduction

Linux is one of the most widely used open-source operating systems in the world and serves as the foundation for a large percentage of servers, cloud platforms, supercomputers, and enterprise systems. Due to its stability, security, flexibility, and efficiency, Linux has become the preferred operating system for software development, system administration, and DevOps practices. Organizations such as Google, Amazon, Facebook, and Netflix rely heavily on Linux-based infrastructure to manage their services and applications. As a result, understanding Linux is considered a fundamental skill for anyone aspiring to build a career in DevOps, Cloud Computing, System Administration, or Software Engineering.

One of the most powerful features of Linux is its Command Line Interface (CLI), which allows users to interact with the operating system by executing commands through a terminal. Unlike graphical user interfaces, the command line provides direct control over the system and enables users to perform tasks more efficiently. Through the terminal, users can navigate directories, create and manage files, monitor system resources, automate repetitive tasks, and configure system settings. The command-line environment is particularly important in DevOps because most servers and cloud instances are managed remotely using terminal-based tools.

The Linux file system is organized in a hierarchical structure consisting of directories and files. Understanding how to navigate this structure and manage resources is essential for effective system administration. Commands such as pwd, ls, cd, mkdir, touch, cat, cp, mv, and rm are commonly used to perform day-to-day operations. These commands form the building blocks for more advanced Linux concepts and are frequently used in professional environments. Learning these commands helps users develop confidence in interacting with Linux systems and prepares them for automation and scripting tasks.

In the field of DevOps, Linux plays a vital role because most automation tools, containerization platforms, cloud services, and deployment environments are designed to work seamlessly with Linux. DevOps engineers regularly use Linux commands to configure servers, deploy applications, monitor system performance, manage logs, and troubleshoot issues. Therefore, mastering the Linux command line is an important first step toward understanding modern software development and deployment workflows.

This project focuses on introducing the fundamental concepts of Linux and command-line operations. Through hands-on practice, various file and directory management tasks are performed using the Linux terminal. Activities such as creating folders, navigating directories, generating files, viewing file contents, copying files, renaming resources, and deleting unnecessary files provide practical exposure to real-world command-line operations. The project was completed using Windows Subsystem for Linux (WSL), which offers a Linux environment within the Windows operating system.

By completing this project, a strong foundation has been established in Linux system navigation and command-line usage. The practical experience gained through these exercises enhances technical proficiency and prepares learners for advanced DevOps tools, shell scripting, automation techniques, and cloud-based infrastructure management. This project serves as the starting point for developing the essential Linux skills required in modern IT and DevOps environments.

Objective

The objective of this project is to gain hands-on experience with basic Linux commands and command-line operations used in DevOps environments. This project focuses on understanding the Linux file system, creating and managing directories and files, navigating through directories, viewing file contents, and performing file operations such as copying, renaming, and deleting. By completing these tasks, I developed a foundational understanding of Linux terminal usage, which is an essential skill for system administration and DevOps practices.

Step 1: Open WSL Terminal

```
jania_jose_kannath@JANIAJO
jania_jose_kannath@JANIAJOSE:~$ whoami
uname -a
jania_jose_kannath
Linux JANIAJOSE 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_6
4 x86_64 GNU/Linux
jania_jose_kannath@JANIAJOSE:~$
```

Step 2: Check Current Directory

```
jania_jose_kannath@JANIAJOSE:~$ pwd
/home/jania_jose_kannath
jania_jose_kannath@JANIAJOSE:~$ ls
1             exp10.c.pdf  exp14        exp8.1.c      exp9.2.c.pdf  reciever.c      shmnd.c      stats
14.c          exp12        exp14.c      exp8.1.c.pdf  exp9.3        reciever.c.pdf  shmrec
C_Programs    exp12.c      exp14.c.pdf  exp9.1        exp9.3.c      s1              shmrec.c
a.out         exp12.c.pdf  exp7         exp9.1.c      exp9.3.c.pdf  s2              shmrec.c.pdf
bankers.pdf   exp13        exp7.c       exp9.1.c.pdf  exp9.4.c      sender          shmsnd
exp10         exp13.c      exp7.c.pdf   exp9.2        exp9.4.c.pdf  sender.c        shmsnd.c
exp10.c       exp13.c.pdf  exp8.1       exp9.2.c      reciever      sender.c.pdf    shmsnd.c.pdf
jania_jose_kannath@JANIAJOSE:~$
```

Step 3: Create Project Folder

```
jania_jose_kannath@JANIAJOSE:~$ mkdir DevOps_Project1
jania_jose_kannath@JANIAJOSE:~$ ls
1                exp10.c.pdf  exp14.c       exp9.1        exp9.3.c.pdf    sender        shmsnd.c
14.c             exp12        exp14.c.pdf   exp9.1.c      exp9.4.c        sender.c      shmsnd.c.pdf
C_Programs       exp12.c      exp7          exp9.1.c.pdf  exp9.4.c.pdf    sender.c.pdf  stats
DevOps_Project1  exp12.c.pdf  exp7.c        exp9.2        reciever        shmnd.c
a.out            exp13        exp7.c.pdf    exp9.2.c      reciever.c      shmrec
bankers.pdf      exp13.c      exp8.1        exp9.2.c.pdf  reciever.c.pdf  shmrec.c
exp10            exp13.c.pdf  exp8.1.c      exp9.3        s1              shmrec.c.pdf
exp10.c          exp14        exp8.1.c.pdf  exp9.3.c      s2              shmsnd
jania_jose_kannath@JANIAJOSE:~$
```

Step 4: Create Subdirectories

```
jania_jose_kannath@JANIAJOSE:~$ cd DevOps_Project1
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ mkdir Documents
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ mkdir Scripts
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ mkdir Logs
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls
Documents  Logs  Scripts
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Step 5: Create Files

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ touch Documents/notes.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ touch Scripts/script.sh
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ touch Logs/log.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Step 6: List Files and Folders

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Documents
notes.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Scripts
script.sh
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Logs
log.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Step 7: Add and View File Content

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ echo "Linux Command Line Basics Project" > Documents/notes.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ cat Documents/notes.txt
Linux Command Line Basics Project
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Step 8: Copy and Rename Files

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ cp Documents/notes.txt Documents/backup_notes.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ mv Scripts/script.sh Scripts/project_script.sh
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Documents
backup_notes.txt  notes.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Scripts
project_script.sh
```

Step 9: Delete a File

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ rm Logs/log.txt
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ ls Logs
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Step 10: Final Directory Structure

```
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$ find .
.
./Documents
./Documents/backup_notes.txt
./Documents/notes.txt
./Logs
./Scripts
./Scripts/project_script.sh
jania_jose_kannath@JANIAJOSE:~/DevOps_Project1$
```

Conclusion

In this project, I successfully practiced and executed fundamental Linux commands using the WSL (Windows Subsystem for Linux) terminal. I learned how to navigate the Linux file system, create and manage directories and files, view and modify file contents, and perform various file management operations. Through this hands-on exercise, I gained practical experience with the Linux command line and strengthened my understanding of core concepts required in DevOps and system administration. This project provided a solid foundation for working with Linux-based environments and prepared me for more advanced DevOps tasks in future projects.